Md. Fazle Rabbi Rithik

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LinkedIn GitHub Codeforces

SUMMARY

Computer Science graduate with strong foundations in object-oriented programming, data structures, algorithms, and analytical problem-solving. Hands-on experience building web applications and REST APIs with Python, FastAPI, JavaScript, SQL, and databases. Developed an automatic dataset-labeling application and a full-stack library management system; also completed thesis research involving YOLOv8 experiments and computer-vision model evaluation. Quick to learn new tools and frameworks, with a focus on readable, maintainable code.

EDUCATION

Rajshahi University of Engineering & Technology (RUET) March 2022 – August 2026
B.Sc. in Computer Science & Engineering

TECHNICAL SKILLS

Languages:	C++ (primary), Python, JavaScript, SQL, C (basic)
Core CS:	Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks
Backend:	FastAPI, Node.js, REST API development
Databases:	PostgreSQL, MongoDB, SQLAlchemy, Mongoose, SQL joins and constraints
Tools:	Git, GitHub, Docker, Docker Compose, Linux, Pytest
Web:	HTML, CSS, JavaScript

PROJECTS

Annotation Wizard (Auto-Labeling) | *FastAPI, ONNX Runtime, PostgreSQL, Docker* Live Demo | GitHub

- Built an automated labeling service that accepts ZIP archives containing image datasets, runs YOLO inference using ONNX Runtime to detect four road-damage classes, and returns YOLO-format annotations in a ZIP archive containing only label files.
- Implemented dataset jobs for human review, bounding-box and class editing, and YOLO-format export. Added API-key-protected endpoints and PostgreSQL persistence, with local services configured using Docker Compose.

Library Management System | *Node.js, Express.js, MongoDB, Mongoose, JavaScript* GitHub

- Built a full-stack application with separate admin/student workflows for book inventory management, browsing, borrowing and returns, and borrowing-history tracking.
- Implemented Express REST endpoints and Mongoose models for users, books, and borrowing records; added user registration/login with bcrypt-based password hashing.

COMPETITIVE PROGRAMMING & ACHIEVEMENTS

- **ICPC Asia Dhaka Regional 2025:** Ranked **73rd** with **Ruet_Potato_Coders**.
- **DUET IUPC 2025:** Ranked **73rd** with **RUET_BRUTEFORCE**.
- **Problem Setter, CodeSmash 2022:** Designed programming problems, statements, and test cases for an ICPC-format RUET fresher contest.
- **RUET Intra-University Contests:** Ranked **6th** in CODESPARK 2022 (team contest) and **6th** in CODESMASH 2021 (individual contest).
- **Codeforces:** Specialist, maximum rating **1422**; solved **1,200+** problems across Codeforces, LeetCode, CSES, and LightOJ.

UNDERGRADUATE THESIS

Class-Aware Data Augmentation for Vision-Based Road Damage Detection: A Systematic Ablation Study

- Designed and ran 8+ YOLOv8 experiments to evaluate geometric, color, blur/noise, and occlusion/mixing augmentation strategies across four road-damage classes.
- Compared class-level and overall detection performance using precision, recall, and mAP50, and authored the thesis documentation.